

Math 484: Nonlinear programming

Chapter 5: Lecture 7

A *posynomial term* is a product $Ct_1^{\alpha_1} \cdots t_k^{\alpha_k}$, where $\alpha_1, \dots, \alpha_k \in \mathbb{R}$ and $C > 0$. A posynomial is a sum of one or more posynomial terms.

A *constrained geometric program* is a problem of the form

$$(GP) \quad \begin{cases} \underset{\mathbf{t} \in (0, \infty)^k}{\text{minimize}} & g(\mathbf{t}) \\ \text{subject to} & g_1(\mathbf{t}) \leq 1, \\ & \dots \\ & g_m(\mathbf{t}) \leq 1, \end{cases}$$

where $g, g_1, \dots, g_m$ are posynomials.

We learn from KKT duality that if GP is super consistent, then GP has a dual.

There exists feasible $\mathbf{t}$ that makes all constraints slack (< 1)

Constrained geometric program dual

Using KKT duality, we can prove that every superconsistent constrained geometric program

$$(GP) \quad \begin{cases} \text{minimize}_{\mathbf{t} \in (0, \infty)^k} & g(\mathbf{t}) \\ \text{subject to} & g_1(\mathbf{t}) \leq 1, \\ & \dots \\ & g_m(\mathbf{t}) \leq 1, \end{cases}$$

has a dual program

$$(DGP) \quad \begin{cases} \text{maximize}_{\delta \geq 0} & v(\delta) \\ \text{subject to} & \dots \end{cases}$$

Maximizing $v(\delta)$ will help us

with variables $\delta = (\delta_1, \dots, \delta_k)$.

minimize $g(\mathbf{t})$

From last time:

$$(GP) \quad \begin{cases} \text{minimize} & \frac{4}{t_1} + \frac{9}{t_2} \\ & t_1, t_2 > 0 \\ \text{subject to} & t_1 + t_2 \leq 1. \end{cases}$$

Dual program.
Can be derived rigorously
from KKT duality.

$$(DGP) \quad \begin{cases} \text{maximize} & v(\delta) \\ & \delta \geq 0 \\ \text{subject to} & \begin{cases} -\delta_1 + \delta_3 = 0 \\ -\delta_2 + \delta_4 = 0 \\ \delta_1 + \delta_2 = 1 \end{cases} \end{cases}$$

Similar to unconstrained

$$v(\delta) = \left(\frac{4}{\delta_1}\right)^{\delta_1} \left(\frac{9}{\delta_2}\right)^{\delta_2} \left(\frac{1}{\delta_3}\right)^{\delta_3} \left(\frac{1}{\delta_4}\right)^{\delta_4}$$

Unfamiliar
not like unconstrained
 $(\delta_3 + \delta_4)^{\delta_3 + \delta_4}$.
GP duality.

Looks a lot like unconstrained dual

Variables

(1)

(Constraints, objective function)

(2)

(3)

The variable in the dual program is $\delta = (\delta_1, \dots, \delta_k)$. $k \neq \#$
The number of coordinates is equal to the number of ^{of primal} posynomial terms in the primal program, including constraint terms.

$$(GP) \quad \begin{cases} \text{minimize} & \frac{4}{t_1} + \frac{9}{t_2} \\ \text{subject to} & t_1 + t_2 \leq 1 \end{cases}$$

Do not count $t_1, t_2 > 0$

1 2
3 4

(do not count ≤ 1 as a term)

We had four posynomials, so we ended up with $\delta = (\delta_1, \delta_2, \delta_3, \delta_4)$.

Constraints

of constraints = ^{obj. func.} 1 + # of vars t_i

- The δ_i 's corresponding to the primal objective function add to 1.
- For each t_i , take the exponent of t_i in each term j , and make it a coeff. of δ_j . These terms add to zero.

$$(GP) \quad \begin{cases} \underset{t_1, t_2 > 0}{\text{minimize}} & 1 \left(\frac{4}{t_1} \right) + \left(\frac{9}{t_2} \right) 2 \\ \text{subject to} & \underset{3}{t_1} + \underset{4}{t_2} \leq 1. \end{cases}$$

Obj func has terms 1, 2.

$$\boxed{\delta_1 + \delta_2 = 1}$$

Related to constraints in KKT dual

$$\begin{aligned} t_1: \text{exp. } -1, 0, 1, 0 & \Rightarrow -\delta_1 + 0\delta_2 + 1\delta_3 + 0\delta_4 = 0 \\ t_2: \text{exp. } 0, -1, 0, 1 & \Rightarrow 0\delta_1 - \delta_2 + 0\delta_3 + 1\delta_4 = 0 \end{aligned}$$

Constraints: Another example

Don't forget

$$\delta_1, \dots, \delta_5 \geq 0.$$

(GP)

minimize

$$t_1, t_2 > 0$$

subject to

$$\frac{1}{t_1^2} + \frac{3}{t_2^3} + t_1 t_2$$

$$t_1 t_2 + \frac{t_2}{2t_1} \leq 1.$$

5 variables

3 constraints

Obj func, t_1, t_2

$$\text{Obj func: } \delta_1 + \delta_2 + \delta_3 = 1$$

$$t_1: -2, 0, 1, 1, 1 \Rightarrow -2\delta_1 + 0\delta_2 + 1\delta_3 + 1\delta_4 - 1\delta_5 = 0$$

$$t_2: 0, -3, 1, 1, 1$$

Dual objective function

- One factor in the dual objective function is $v_0(\delta)$, the **unconstrained geometric program dual**.
- For each constraint with terms $i, \dots, j$, we have a factor $(\delta_i + \dots + \delta_j)^{\delta_i + \dots + \delta_j}$ in the dual objective function.

$$(GP) \quad \begin{cases} \text{minimize}_{t_1, t_2 > 0} & \frac{4}{t_1} + \frac{9}{t_2} \\ \text{subject to} & t_1 + t_2 \leq 1. \end{cases}$$

Imagine $\tilde{g}(t_1, t_2) = \frac{4}{t_1} + \frac{9}{t_2} + t_1 + t_2$
as obj func in unconstrained GP.

Then, $v_0(\delta)$ is dual program to $\tilde{g}(t_1, t_2)$

Here, $v_0(\delta) = \left(\frac{4}{\delta_1}\right)^{\delta_1} \left(\frac{9}{\delta_2}\right)^{\delta_2} \left(\frac{1}{\delta_3}\right)^{\delta_3} \left(\frac{1}{\delta_4}\right)^{\delta_4}$

In other words, copy constant coeffs C_i ,
get factors $\left(\frac{C_i}{\delta_i}\right)^{\delta_i}$, one for each
primal term.

We have constraint with terms 3, 4.

This gives dual obj func factor
 $(\delta_3 + \delta_4)^{\delta_3 + \delta_4}$.

Dual obj func is $v(\delta) = \left(\frac{4}{\delta_1}\right)^{\delta_1} \left(\frac{9}{\delta_2}\right)^{\delta_2} \left(\frac{1}{\delta_3}\right)^{\delta_3} \left(\frac{1}{\delta_4}\right)^{\delta_4} \underbrace{(\delta_3 + \delta_4)^{\delta_3 + \delta_4}}_{\text{One factor like this for each constraint}}$

Something I forgot to say
during class:

S_i can equal zero.

In this case, we interpret

$$\begin{aligned}\left(\frac{C_i}{S_i}\right)^{S_i} &= \left(\frac{C_i}{0}\right)^0 \\ &= \frac{C_i^0}{0^0} \\ &= \frac{1}{1} = 1\end{aligned}$$

using the "convention" $0^0 = 1$.

This almost never is an issue. (and will not appear on exams)

Misha's notes cover this caveat
in more detail.

Dual objective function: another example

$$(GP) \quad \begin{cases} \text{minimize}_{t_1, t_2 > 0} & \overset{1}{\frac{1}{t_1^2}} + \overset{2}{\frac{3}{t_2^3}} + \overset{3}{t_1 t_2} \\ \text{subject to} & \overset{4}{t_1 t_2} + \overset{5}{\frac{t_2}{2t_1}} \leq 1 \\ & \overset{6}{t_1} + \overset{7}{\frac{t_2}{2t_1^4}} \leq 1. \end{cases}$$

Variables $\delta_1, \dots, \delta_7$.

2 constraints $\Rightarrow$ 2 factors of form $(\delta_i + \dots + \delta_j)$

(1) $\cdot (\delta_4 + \delta_5)^{\delta_4 + \delta_5}$

(Constraint 1, Terms 4,5)

(2) $\cdot (\delta_6 + \delta_7)^{\delta_6 + \delta_7}$

(Constraint 2, 6,7)

(3) (Unconstrained dual)

$$V_0(\delta) = \left(\frac{1}{\delta_1}\right)^{\delta_1} \left(\frac{3}{\delta_2}\right)^{\delta_2} \left(\frac{1}{\delta_3}\right)^{\delta_3} \left(\frac{1}{\delta_4}\right)^{\delta_4} \left(\frac{1}{2}\right)^{\delta_5} \left(-\frac{1}{2}\right)^{\delta_6} \left(-\frac{1}{2}\right)^{\delta_7}$$

Final dual obj func is

$$V(\delta) = V_0(\delta) (\delta_4 + \delta_5)^{\delta_4 + \delta_5} (\delta_6 + \delta_7)^{\delta_6 + \delta_7}$$

Conditions at solution

- By KKT duality, $g(\mathbf{t}^*) = v(\delta^*)$. *At solutions, primal and dual agree.*
- Each primal term has a solution condition.
- For each **objective function** posynomial $u_i(\mathbf{t})$, we have

$$\underbrace{v(\delta^*)} = \underbrace{g(\mathbf{t}^*)} = \frac{u_i(\mathbf{t}^*)}{\delta_i^*} = \frac{\text{ith term}}{\delta_i^*}$$

- For each **constraint term** $u_\ell(\mathbf{t})$, we have

$$u_\ell(\mathbf{t}^*) = \frac{\delta_\ell^*}{\delta_i^* + \dots + \delta_j^*},$$

If primal term ℓ is in a constraint with terms $i, \dots, j$, then $\frac{\delta_\ell^}{\delta_i^* + \dots + \delta_j^*}$*

where the constraint containing $u_\ell(\mathbf{t})$ consists of terms $i, \dots, j$.

is equal to val of ℓ th term at solution

Something I forgot to say in class:

It is possible that (GP) has a constraint with terms $i, i+1, \dots, j-1, j$, but at the dual solution,

$$\delta_i^* = \delta_{i+1}^* = \dots = \delta_{j-1}^* = \delta_j^* = 0.$$

In this case, the condition

$$u_\ell(t^*) = \frac{\delta_\ell^*}{\delta_i^* + \dots + \delta_j^*} = \frac{0}{0} \quad (\text{with } i \leq \ell \leq j)$$

does not make sense, and we ignore this meaningless condition.

Conditions at solution

$$(GP) \quad \begin{cases} \underset{t_1, t_2 > 0}{\text{minimize}} & \frac{4}{t_1} + \frac{9}{t_2} \\ \text{subject to} & t_1 + t_2 \leq 1. \end{cases}$$

- We had $\delta^* = (\frac{2}{5}, \frac{3}{5}, \frac{2}{5}, \frac{3}{5})$, $\mathbf{t}^* = (\frac{2}{5}, \frac{3}{5})$. We compute

Check conditions

$$v(\delta^*) = 25.$$

Obj. func. conditions:

$$\frac{u_1(t_1^*, t_2^*)}{s_1^*} = v(\delta^*)$$

$$\Rightarrow \frac{4/t_1^*}{2/5} = 25 \Leftrightarrow t_1^* = \frac{2}{5}$$



Constraint conditions

$$u_3(t^*) = \frac{\delta_3^*}{\delta_3^* + \delta_4^*} \quad \text{Term 3}$$

In constraint with terms 3, 4.

$$t_i^* = \frac{2/5}{2/5 + 3/5} = \frac{2}{5} \quad \checkmark$$

If we had constraint with terms 5, 6, 7 then we have condition

$$u_5(t^*) = \frac{\delta_5^*}{\delta_5^* + \delta_6^* + \delta_7^*}$$

A full example

$$(GP) \quad \begin{cases} \underset{t_1, t_2, t_3 > 0}{\text{minimize}} & \frac{40}{t_1 t_2 t_3} + 40 t_2 t_3 \\ \text{subject to} & \frac{1}{2} t_1 t_3 + \frac{1}{4} t_1 t_2 \leq 1. \end{cases}$$

A full example

$$(GP) \quad \begin{cases} \text{minimize}_{t_1, t_2, t_3 > 0} & \frac{40}{t_1 t_2 t_3} + 40 t_2 t_3 \\ \text{subject to} & \frac{1}{2} t_1 t_3 + \frac{1}{4} t_1 t_2 \leq 1. \end{cases}$$

The dual will have variables $\delta = (\delta_1, \dots, \delta_4)$.

$$\delta_1 + \delta_2 = 1.$$

$$-\delta_1 + \delta_3 + \delta_4 = 0.$$

$$-\delta_1 + \delta_2 + \delta_4 = 0.$$

$$-\delta_1 + \delta_3 + \delta_3 = 0.$$

$$\left\{ \begin{array}{ll} \text{maximize}_{\delta \geq 0} & \left(\frac{40}{\delta_1}\right)^{\delta_1} \left(\frac{40}{\delta_2}\right)^{\delta_2} \left(\frac{1}{2\delta_3}\right)^{\delta_3} \left(\frac{1}{4\delta_2}\right)^{\delta_4} \\ & \cdot (\delta_3 + \delta_4)^{\delta_3 + \delta_4} \\ \text{subject to} & \delta_1 + \delta_2 = 1 \\ & -\delta_1 + \delta_3 + \delta_4 = 0 \\ & -\delta_1 + \delta_2 + \delta_4 = 0 \\ & -\delta_1 + \delta_3 + \delta_3 = 0 \end{array} \right.$$

One
feasible

δ

$= \delta^*$

$$\left\{ \begin{array}{ll} \text{maximize}_{\delta \geq 0} & \left(\frac{40}{\delta_1}\right)^{\delta_1} \left(\frac{40}{\delta_2}\right)^{\delta_2} \left(\frac{1}{2\delta_3}\right)^{\delta_3} \left(\frac{1}{4\delta_2}\right)^{\delta_4} \\ & \cdot (\delta_3 + \delta_4)^{\delta_3 + \delta_4} \\ \text{subject to} & \delta_1 + \delta_2 = 1 \\ & -\delta_1 + \delta_3 + \delta_4 = 0 \\ & -\delta_1 + \delta_2 + \delta_4 = 0 \\ & -\delta_1 + \delta_3 + \delta_3 = 0 \end{array} \right.$$

$$\delta^* = \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right), \quad \underline{v(\delta^*) = 60.}$$

$$\delta^* = (\frac{2}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3})$$

$$(GP) \quad \begin{cases} \text{minimize}_{t_1, t_2, t_3 > 0} & \frac{40}{t_1 t_2 t_3} + 40 t_2 t_3 \\ \text{subject to} & \frac{1}{2} t_1^3 + \frac{1}{4} t_1 t_2^4 \leq 1. \end{cases}$$

1st term
 δ_1^*

$$60 = v(\delta^*) = \frac{40}{\frac{2}{3} t_1 t_2 t_3} = \frac{40 t_2 t_3}{\frac{1}{3}}$$

3rd term
 $= \frac{\delta_3^*}{\delta_3^* + \delta_4^*}$

$$\frac{1}{2} t_1 t_3 = \frac{1/3}{1/3 + 1/3} = \frac{1}{2}$$

$$\frac{1}{4} t_1 t_2 = \frac{1/3}{1/3 + 1/3} = \frac{1}{2}$$

$$60 = \frac{40}{\frac{2}{3}t_1 t_2 t_3} = \frac{40t_2 t_3}{\frac{1}{3}}.$$

$$\frac{1}{2}t_1 t_3 = \frac{1}{2}.$$

$$\frac{1}{4}t_1 t_2 = \frac{1}{2}.$$

This becomes a linear system with variables $\log t_1, \log t_2, \log t_3$. Solving, we have

$$(t_1^*, t_2^*, t_3^*) = \left(2, 1, \frac{1}{2}\right). \quad \checkmark$$

Systematic.